

System Data Dictionary

Restaurant Management Platform

Database Specification

August 10, 2026

1 Table Specifications

1.1 Table: users

Field Name	Data Type	Key / Constraints	Description
id	BIGINT	PK, Auto-increment	Primary key identifier for the user.
name	VARCHAR(255)	NOT NULL	User's full name.
email	VARCHAR(255)	UNIQUE, NOT NULL	Account email address used for logging in.
password	VARCHAR(255)	NOT NULL	Encrypted password credential.

1.2 Table: owners

Field Name	Data Type	Key / Constraints	Description
user_id	BIGINT	PK, FK → users.id	Primary key referencing the parent user.
business_tax_id	VARCHAR(50)	NULLABLE	Tax/EIN identification number.
phone_number	VARCHAR(20)	NULLABLE	Contact telephone number for business operations.
is_identity_verified	BOOLEAN	DEFAULT false	KYC / identity verification flag.

1.3 Table: customers

Field Name	Data Type	Key / Constraints	Description
<code>user_id</code>	BIGINT	PK, FK → <code>users.id</code>	Primary key referencing the parent user.
<code>phone_number</code>	VARCHAR(20)	NULLABLE	Personal contact phone number.
<code>date_of_birth</code>	DATE	NULLABLE	Date of birth for promotion targeting.
<code>loyalty_points</code>	INTEGER	DEFAULT 0	Accumulated diner rewards points.
<code>dietary_preferences</code>	TEXT	NULLABLE	Preferences (e.g., Vegan, Gluten-Free, Allergies).

1.4 Table: restaurants

Field Name	Data Type	Key / Constraints	Description
<code>id</code>	BIGINT	PK, Auto-increment	Primary key for the restaurant.
<code>owner_id</code>	BIGINT	FK → <code>owners.user_id</code>	Foreign key pointing to the managing owner.
<code>name</code>	VARCHAR(255)	NOT NULL	Public name of the establishment.
<code>description</code>	TEXT	NULLABLE	Public overview or description of the restaurant.
<code>address</code>	VARCHAR(255)	NOT NULL	Street address location.
<code>cuisine_type</code>	VARCHAR(100)	NOT NULL	Primary type of cuisine offered.
<code>price_range</code>	ENUM	<code>price_range_enum</code>	Price rating category ('\$' to '\$\$\$\$').
<code>avg_rating</code>	DECIMAL(2,1)	DEFAULT 0.0	Calculated average customer review rating.

1.5 Table: categories

Field Name	Data Type	Key / Constraints	Description
<code>id</code>	BIGINT	PK, Auto-increment	Primary key for the menu category.
<code>restaurant_id</code>	BIGINT	FK → <code>restaurants.id</code>	Foreign key pointing to the associated restaurant.
<code>name</code>	VARCHAR(100)	NOT NULL	Name of category (e.g., Starters, Mains, Drinks).

1.6 Table: menu_items

Field Name	Data Type	Key / Constraints	Description
<code>id</code>	BIGINT	PK, Auto-increment	Primary key for the menu item.
<code>category_id</code>	BIGINT	FK → <code>categories.id</code>	Foreign key identifying the parent category.
<code>name</code>	VARCHAR(255)	NOT NULL	Item name.
<code>description</code>	TEXT	NULLABLE	Item description, ingredients, or allergens.
<code>price</code>	DECIMAL(8,2)	NOT NULL	Cost of the item.
<code>is_available</code>	BOOLEAN	DEFAULT <code>true</code>	Toggle indicating current item availability.

1.7 Table: reviews

Field Name	Data Type	Key / Constraints	Description
<code>id</code>	BIGINT	PK, Auto-increment	Primary key for the review entry.
<code>restaurant_id</code>	BIGINT	FK → <code>restaurants.id</code>	Foreign key targeting the reviewed restaurant.
<code>customer_id</code>	BIGINT	FK → <code>customers.user_id</code>	Foreign key referencing the reviewing customer.
<code>rating</code>	SMALLINT	CHECK (1–5)	Numerical rating score from 1 to 5 stars.
<code>comment</code>	TEXT	NULLABLE	Written feedback text.

1.8 Table: review_replies

Field Name	Data Type	Key / Constraints	Description
<code>id</code>	BIGINT	PK, Auto-increment	Primary key for the reply.
<code>review_id</code>	BIGINT	FK → <code>reviews.id</code>	Foreign key pointing to the target customer review.
<code>owner_reply</code>	TEXT	NOT NULL	Response text written by the owner.

1.9 Table: dining_logs

Field Name	Data Type	Key / Constraints	Description
<code>id</code>	BIGINT	PK, Auto-increment	Primary key for the log record.
<code>customer_id</code>	BIGINT	FK → <code>customers.user_id</code>	Foreign key referencing the customer.
<code>restaurant_id</code>	BIGINT	FK → <code>restaurants.id</code> , NULL	Optional link to a system restaurant entity.
<code>amount_spent</code>	DECIMAL(8,2)	NOT NULL	Total money spent on the transaction.
<code>note</code>	VARCHAR(255)	NULLABLE	Personal memo or note recorded by the customer.